SPECX · TEAM WORK BREAKDOWN · 14-WEEK SEMESTER PLAN

Work Split Across Three Devs

Split by pipeline layer and tech stack, so each dev owns one coherent slice and the S6 → S1 → S2 training-data chain stays with one person. The draft's dependency order S6 → S1 → S2 → S4 is preserved.

DEV A · PLATFORM

Owns F3 generation · F1 sandbox & contracts · CLI/CI · reproducibility

Stack TypeScript/Node, Playwright + CDP, Docker, mitmproxy, Postgres

A1	wk 1	Monorepo + CI skeleton; publish the shared interface schemas.
A2	wk 1–2	Two seed apps (e-commerce, CRUD/admin), dockerised with seeded Postgres and deterministic reset.
A3	wk 1–2	F3 generator v0: tree-sitter repo parser, DOM/ARIA analyser, API-trace miner, atomic tool emitter with provenance (S7 retrieval as internal aid).
A4	wk 2	Sandbox skeleton (Playwright + CDP in Docker, mitmproxy). *Exit: a tool run replays deterministically.*
A5	wk 3–4	F1 contract deriver over all five evidence surfaces, effect differ, quarantine manager with machine-readable reasons; expose event stream for S6.
A6	wk 5–7	Sandbox throughput for S1 (parallel sandboxes, fast reset, concurrent calls); third seed app; MCP-B Chrome target as spec-drift fallback.
A7	wk 5–7	Typed workflow graph builder + manifest exporter, ready by week 7 for S2.
A8	wk 8–11	Wire S6, S1, S2 into the verify stage; CLI (generate / verify / optimise / evaluate); GitHub Actions.
A9	wk 12–13	Keep benchmark infra stable; freeze Chrome version and pin the spec.
A10	wk 14	One-command reproducibility package from a clean clone.

DEV B · SAFETY ML (S6 → S1 → S2)

Owns S6 anomaly layer · S1 DRL fuzzer · S2 GNN risk model

Stack Python, scikit-learn, PyTorch, Gymnasium/SB3, PyTorch Geometric

B1	wk 1–2	Feature windower over network + MutationObserver streams (recorded data until A4 lands); Gym env design against a mocked sandbox; random-fuzzer baseline.
B2	wk 3–4	S6: Isolation Forest + LSTM autoencoder on approved traces; injected slow-leak set (delayed POSTs, storage writes, third-party calls). *Exit: leaks detected, benign false-positive rate measured.*
B3	wk 5–7	S1: PPO/DQN on the real sandbox, violation reward + curiosity bonus; coverage-guided baseline; log every verdict in S2's label format. *Exit: more violations per sandbox-hour than baselines.*
B4	wk 8–9	S2: GCN/GAT with edge-prediction + risk-classification heads; rule-based and MLP baselines. *Exit: beats rule-based checker on held-out precision/recall.*
B5	wk 10–13	Ablations for S1, S2 and S6 using C's stats engine (no second copy). Threats to validity: S1 reward hacking, realism of S6's injected leaks.

DEV C · OPTIMISATION & EVALUATION

Owns F2 loop · S4 patch model · F4 benchmark · stats engine · dashboard

Stack Python, LLM/VLM APIs, PEFT (LoRA), SciPy, React

C1	wk 1–2	Evaluation agent harness: frozen agent, WebArena/Mind2Web tasks adapted to the seed apps, held-out split; success, steps/tokens, policy-violation metrics.
C2	wk 2–4	Shared stats engine: paired trials, effect sizes, CIs, Holm–Bonferroni. Run raw-DOM and generated baselines early.
C3	wk 5–7	F2 closed loop with text-trace LLM revisions + regression gate (accept only if performance improves and no safety invariant regresses). Doubles as S4's baseline.
C4	wk 5–9	S4 dataset: failure→patch pairs from F2 runs + synthetic failure injection (mutated schemas, broken selectors).
C5	wk 8–11	S4: LoRA fine-tune of a Qwen-VL/LLaVA-class model, schema-constrained JSON, plugged into F2; run both S4 ablations. *Exit: patches apply cleanly; fewer iterations than text baseline.*
C6	wk 12–13	F4 four-condition benchmark, ≥3 apps, ≥10 paired trials per condition; agent-readiness reports.
C7	wk 6–14	React dashboard: readiness reports, quarantine list, revision log, scores.

Week 1 · shared interfaces

Tool definition + provenance record	A → B, C
Sandbox API: *reset(seed)*, *invoke*, *invoke_concurrent* → verdict + effect diff (language-neutral JSON: A is TypeScript, B/C are Python)	A → B, C
Event-stream record (network + DOM mutations)	A → B
Verdict / quarantine record (also S2's training labels)	A → B
Typed workflow graph schema	A → B, C
Trial-result record + stats engine	C → B

Decision gates

End wk 2 · Sandbox replay works; B and C move from mocks to real data.

End wk 7 · S1 go or fallback. Fallback promotes S7 (RAG test generation) to the fourth contribution, which lands on **A** (S7 lives in F3).

End wk 9 · S2 go or demote; check S4 has enough training pairs.

End wk 11 · Feature freeze.

Wk 12 · Freeze agent model + Chrome version; record seeds.

Paper split

A · System (C1); related work vs ContractSkill, CoEvoSkills, formal verification; artifact; final prior-art re-run.

B · C2, C3, C5 (S1, S2, S6): methods + ablations.

C · C4 (S4), C6 (benchmark), experimental design, statistics, threats to validity.

Red-team rotation · A checks B's novelty claims, B checks C's, C checks A's.

Changes from the draft

F2 moves from wk 10–11 to wk 5–7. Keeps C busy early and yields more S4 training pairs (addresses the draft's sparse-pairs risk).

Workflow graph builder goes to A, due wk 7. One owner for a schema both S2 and F4 use.

Biggest risk: A's sandbox in weeks 1–4. If it slips, everyone slips.